$SrTiO_3$: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

Unless a source is explicitly named, numerical values are hypothetical model assumptions rather than measured device specifications. Show units and state which inference depends on each assumption.

E01 An impedance time scale

An ideal parallel circuit has $R = 1\,\mathrm{M\Omega}$ and $C = 1\,\mu\mathrm{F}$. Derive its imaginary impedance and determine τ and its positive-frequency loss-peak frequency. Does observing this peak uniquely establish the microscopic mechanism?

E02 Charged vacancies and slab resistance

Assume vacancies are the sole carriers, $c_V = 10^{18}\,\mathrm{cm^{-3}}$, $\mu_V = 8.6 \times 10^{-14}\,\mathrm{cm^2/(V\,s)}$, charge $+2q$, $q = 1.602 \times 10^{-19}\,\mathrm{C}$. Calculate σ_V and the ideal bulk resistance of a slab of thickness $0.005\,\mathrm{cm}$ and area $1\,\mathrm{cm^2}$. Define mobility and give two limitations.

E03 A conditional contact-barrier comparison

At $300\,\mathrm{K}$ take $k_B T = 0.02585\,\mathrm{eV}$. Assume two thermionic-emission states have identical effective area, ideality factor, temperature and effective bias; their barrier heights are 0.72 and $0.50\,\mathrm{eV}$. Compute the lower-barrier to higher-barrier current ratio. Why may this differ from a reported device's switching ratio?

E04 Two terminal voltages

For $V_{device} = V_{Ni} - V_{Ti}$, calculate V_{device} when $V_{Ni} = +2\,\mathrm{V}$, $V_{Ti} = -1\,\mathrm{V}$ and when both are $+2\,\mathrm{V}$. Why does a peak voltage alone not establish the conductance update?

E05 A finite device sample

In one stated experiment, 16 devices of 40 met a specified repeatability criterion. Calculate the observed fraction. Give two reasons it is not a universal device yield or a precise probability for other samples.

E06 Pulse-count reduction

Suppose a learning protocol needs 62 identical pulses to reach a stated criterion, while relearning after a specified delay needs 16. Compute the reduction in pulse count relative to the first trial. Under what conditions is this the same percentage reduction in pulse-train duration?

E07 Tracer diffusion and conductivity

In a hypothetical dilute independent-carrier model, $D^* = 6.9 \times 10^{-12}\,\mathrm{cm^2/s}$, $H = 0.69$, $c = 10^{18}\,\mathrm{cm^{-3}}$, $z = 2$, $T = 700\,\mathrm{K}$, $q = 1.602 \times 10^{-19}\,\mathrm{C}$ and $k_B = 1.381 \times 10^{-23}\,\mathrm{J/K}$. Set $D_\sigma = D^*/H$ and $\sigma = D_\sigma c (zq)^2/(k_B T)$. Calculate both. Does a tracer profile alone establish c or a unique microscopic mechanism?

$SrTiO_3$: Exercises (continued)

E08 Sweep time and relaxation

For an illustrative sweep from 0 to 1 V at constant rate, calculate duration at $0.05\,\mathrm{mV/s}$ and at $25\,\mathrm{V/s}$. If a modeled internal relaxation time is 1 s, compare time scales. Explain why the faster sweep cannot automatically be called nondisturbing.

E09 Impedance and microscopic hypotheses

An STO device exhibits I–V hysteresis and a bias-dependent impedance loop. Consider both interface trapping and oxygen migration as competing explanations. Propose complementary controls that could distinguish an interface barrier, bulk transport and slow defect redistribution; explain an alternative origin for the loop and an observation needed to support retention.

E10 A timing-dependent update experiment

Define a sign convention for Δt in a two-terminal pulse-overlap experiment and describe how to determine the actual device voltage versus time. Explain why initial-state preparation, read protocol and the normalization denominator matter when comparing conductance updates.

E11 Film thickness and impedance

Two hypothetical uniform films share area, conductivity and permittivity; their thicknesses are 100 and 200 nm. Derive the ratios of ideal bulk resistance, geometric capacitance and RC. Give two reasons these ratios alone cannot exclude an interface contribution.

E12 Activation-energy inference

An ideal activated model is $\sigma T = C\exp[-E_a/(k_B T)]$ over a limited temperature range. Derive the slope of $\ln(\sigma T)$ against $1/T$, and calculate E_a for slope $-9.28\times10^3\,\mathrm{K}$ using $k_B = 8.617\times10^{-5}\,\mathrm{eV/K}$. Why does a slope change not independently prove that one particular carrier became dominant?

E13 A check circuit for associative learning

In a hypothetical series divider, a $10\,\mathrm{k\Omega}$ resistor and a memory element are driven by a $0.3\,\mathrm{V}$ check pulse; the output is measured across the resistor. Find the output for memory resistances of $90\,\mathrm{k\Omega}$ and $50\,\mathrm{k\Omega}$ and classify each using a strict $V_{out} > 45\,\mathrm{mV}$ criterion. Explain why the calculation alone cannot establish the number of training trials, a non-disturbing read, or biological learning.

E14 Interpret a reported sequence

A particular Ni/Nb:STO/Ti demonstration used CS leading US by 4 ms and a 0.3 V, 1 ms check pulse. The initial response was about 5 mV; after three paired training trials it exceeded the 45 mV criterion. Following 30 paired trials, ten CS-only trials brought the response below that criterion. Identify the training, checking and extinction operations. What controls would help determine whether the check pulse itself affects the state, and why must the trial counts not be treated as universal?